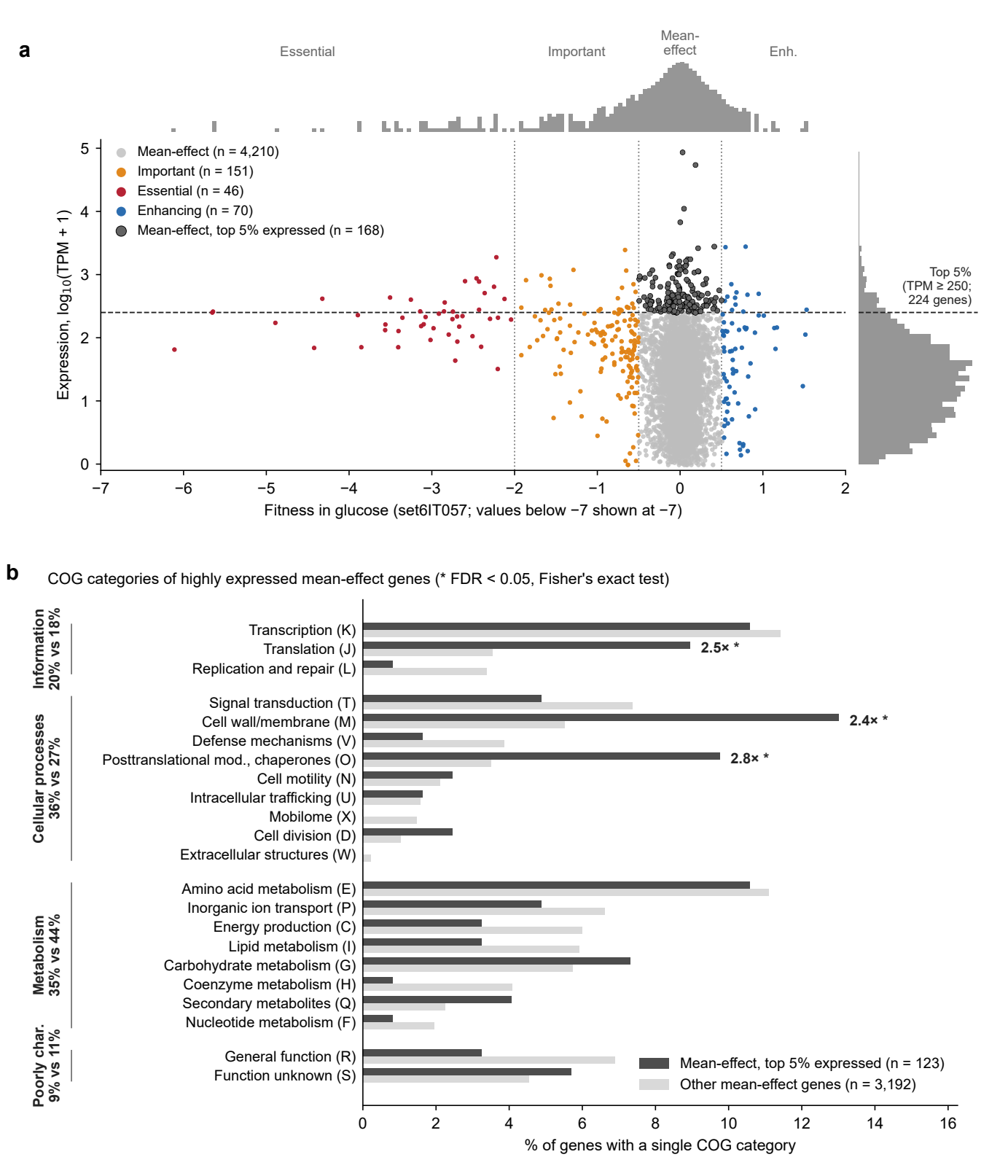

a
Essential
Important
Mean-effect
Enh.
Mean-effect (n = 4,210)
Important (n = 151)
Essential (n = 46)
Enhancing (n = 70)
Mean-effect, top 5% expressed (n = 168)
Expression, log10(TPM + 1)
Top 5% (TPM ≥ 250; 224 genes)
Fitness in glucose (set6IT057; values below −7 shown at −7)
b
COG categories of highly expressed mean-effect genes (* FDR < 0.05, Fisher's exact test)
Information 20% vs 18%
Cellular processes 36% vs 27%
Metabolism 35% vs 44%
Poorly char. 9% vs 11%
Transcription (K)
Translation (J)
Replication and repair (L)
Signal transduction (T)
Cell wall/membrane (M)
Defense mechanisms (V)
Posttranslational mod., chaperones (O)
Cell motility (N)
Intracellular trafficking (U)
Mobilome (X)
Cell division (D)
Extracellular structures (W)
Amino acid metabolism (E)
Inorganic ion transport (P)
Energy production (C)
Lipid metabolism (I)
Carbohydrate metabolism (G)
Coenzyme metabolism (H)
Secondary metabolites (Q)
Nucleotide metabolism (F)
General function (R)
Function unknown (S)
2.5× *
2.4× *
2.8× *
Mean-effect, top 5% expressed (n = 123)
Other mean-effect genes (n = 3,192)
% of genes with a single COG category